ALGEBRA 2 · UNIT 7: RATIONAL FUNCTIONS

Mural Business

Run a mural-painting business with rational functions, work rates, and rational equations.

CP1

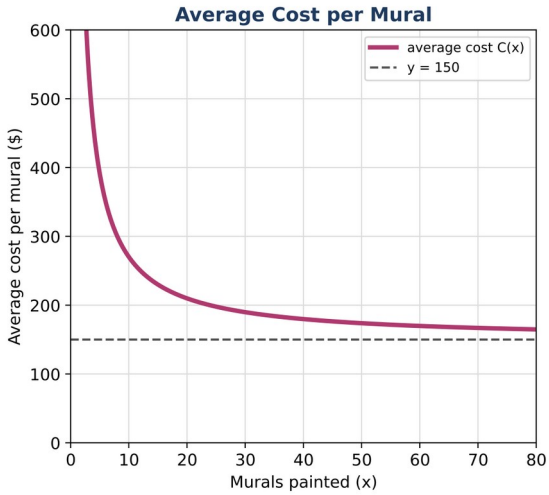
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You and your friends are starting a mural business. Equipment costs $1,200 once, and materials cost $150 per mural. The more murals you paint, the lower your average cost per mural.

You will also schedule painters who work at different speeds and plan your travel. Every question leads to a rational function or equation.



Mural Co.

Equipment $1,200

Materials $150

per mural

Ana: 6 h per mural

Ben: 9 h per mural

Learning Goals

- I can simplify rational expressions and identify restrictions.
- I can solve rational equations and check for extraneous solutions.
- I can interpret asymptotes in context.

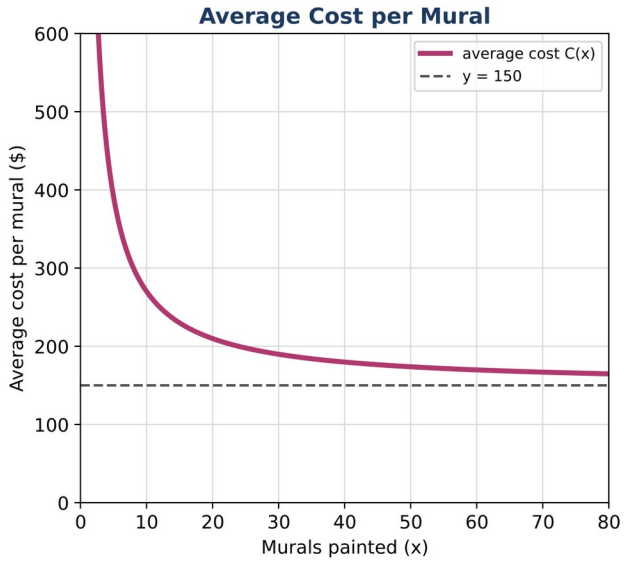
Project Timeline

When	What you will do
Day 1	Average Cost
Day 2	Rational Equations
Day 3	Business Plan
Final	Business Plan + reflection

Part 1: Average Cost

Day 1 · $C(x) = (1200 + 150x) \div x$.

1. Rewrite $C(x)$ as $150 + 1200 \div x$. Identify the vertical and horizontal asymptotes and explain them in context.



Mural Co.

Equipment $1,200

Materials $150

per mural

Ana: 6 h per mural

Ben: 9 h per mural

2. How many murals give an average cost of $175?

3. A larger studio has $C_2(x) = (3000 + 120x) \div x$. For how many murals is the larger studio cheaper per mural?

Part 2: Rational Equations

Day 2 · Check every solution.

4. Ana takes 6 h, Ben 9 h. Together with Cara they take 2.4 h. Find Cara's time with a rational equation.

5. You drive 60 miles to a job and back. You drive 10 mph faster coming home, and the round trip takes 2.7 hours of driving. Find your speed going.

6. Simplify $(x^2 - 9) \div (x^2 + 5x + 6)$. State the restrictions and where the graph has a hole.

Part 3: Business Plan

Day 3 · Plan the season.

7. Set a price and find the break-even number of murals with a rational equation.

8. Schedule a three-person team and find their combined time.

9. Write the plan with a graph.

Final Product: Business Plan

Your final product must include:

- Average cost analysis with asymptotes
- Studio comparison
- Solved rational equations
- Business plan

Reflection

R1. What does the horizontal asymptote mean for the business?

R2. How do you know when a solution is extraneous?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Prices and schedules are based on correctly solved rational equations, with extraneous solutions rejected. Goes beyond the minimum with a creative, realistic choice.	Prices and schedules are based on correctly solved rational equations, with extraneous solutions rejected.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes the average cost function with asymptotes, work-rate schedules, and a pricing decision supported by equations. Polished, organized, and visually engaging.	Plan includes the average cost function with asymptotes, work-rate schedules, and a pricing decision supported by equations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.